ASSIGNMENT 1

Deadline: 11:59 pm, February 1, 2026

Submit via Blackboard
Please follow the course policy and the school's academic honesty policy.

Notations: Boldface character $\mathbf{x}$ denotes a vector and regular character x denotes a scalar.

1 Compulsory Questions

Question 1 (25 %) (Compulsory)

Consider a single update step of the perceptron learning algorithm. You are given the following data point and current weight vector:

- Data point: $\mathbf{x} = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$, $y = +1$.

- Current weight vector: $\mathbf{w} = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$.

1. **Please calculate the prediction $\hat{y}$ for this data point using the formula:**

$$\hat{y} = h(\mathbf{x}) = \mathbf{sign}(\mathbf{w}^T\mathbf{x}).$$

2. **Please determine whether the data point is misclassified.**

3. **If it is misclassified, update the weight vector using the perceptron rule:**

$$\mathbf{w} \leftarrow \mathbf{w} + y\mathbf{x}.$$

Please provide the updated weight vector $\mathbf{w}$.

Question 2 (25 %) (Compulsory)

Consider a learning problem where there are k possible hypotheses $h_1, h_2, \ldots, h_k$, and you have a sample of N independent and identically distributed (i.i.d.) examples. Let μ_i be the true probability that hypothesis h_i makes a correct prediction on an example, and v_i be the fraction of correct predictions made by hypothesis h_i on the sample. We are interested in applying Hoeffding's inequality to bound the probability that v_i deviates from μ_i by more than ϵ.

The Hoeffding's inequality for a single hypothesis is:

$$\mathbb{P}[|v_i - \mu_i| > \epsilon] \leq 2e^{-2\epsilon^2 N}$$

where ϵ is a positive deviation threshold, v_i is the observed fraction of correct predictions for hypothesis h_i, and μ_i is the true probability for h_i.

Now, consider the following:

- **(a) Using Hoeffding's inequality, please calculate the probability that the error of hypothesis h_1 deviates from the true probability μ_1 by more than $\epsilon = 0.05$, given that the sample size is $N = 500$.**

- **(b) Suppose we have $k = 5$ hypotheses, and we want to ensure that for each hypothesis h_i, the deviation is bounded by $\epsilon = 0.05$ with probability at least $1 - \delta$. What sample size N is required for $\delta = 0.01$. Please use Hoeffding's inequality to derive the minimum value of N.**

- **(c) Consider a situation where we have $k = 5$ hypotheses, and you want to ensure that with probability at least $1 - \delta$, the maximum deviation over all hypotheses is bounded by $\epsilon = 0.05$. What sample size N is required to achieve this, given $\delta = 0.01$? Please provide an explanation for why this differs from part (b).**

Question 1

(1) $\vec{y} = h(x) = \text{sign}(w^T x)$

$$= \text{sign}\left(\begin{bmatrix} 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}\right)$$

$$= \text{sign}(-3)$$

$$= -1$$

(2) $\because y = +1 \neq \vec{y} = -1$

$\therefore$ It is misclassified

(3) $w \leftarrow w + yx$

Let w' be the new w.

$$w' \leftarrow w + yx$$

$$yx = +1 \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$$

$$w' = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

Question 2

(a) $P[|v_i - \mu_i| > \varepsilon] \leq 2e^{-2\varepsilon^2 N}$

$$\leq 2e^{-2(0.05)^2(500)}$$

$$P[|v_1 - \mu_1| > 0.05] = 2e^{-2.5}$$

$$\approx 0.16417$$

(b) $2e^{-2\varepsilon^2 N} \leq \delta$

$$e^{-2\varepsilon^2 N} \leq \frac{\delta}{2}$$

$$-2\varepsilon^2 N \leq \ln\frac{\delta}{2}$$

$$\varepsilon^2 N \geq \frac{1}{2}\ln\frac{2}{\delta}$$

$$N \geq \frac{1}{2\varepsilon^2}\ln\frac{2}{\delta}$$

$$N \geq \frac{1}{2(0.05)^2}\ln\frac{2}{0.01}$$

$$N \geq 1059.66$$

$\therefore N_{min} = 1060$

(c) $P[\max_i |v_i - \mu_i| > \varepsilon] \leq 2ke^{-2\varepsilon^2 N}$

$\therefore 2ke^{-2\varepsilon^2 N} \leq \delta$

$$N \geq \frac{1}{2\varepsilon^2}\ln\frac{2k}{\delta}$$

$$N \geq \frac{1}{2(0.05)^2}\ln\frac{2(5)}{(0.01)}$$

$$N \geq 1381.55$$

$\therefore N_{min} = 1382$

We need more sample size because we need to bound to hold simultaneously for all 5 hypothesis, not one at a time

Attachment

Declaration for written assignment

I am/we are submitting the assignment for:
☑ an individual project or
□ a group project on behalf of all members of the group. It is hereby confirmed that the submission is authorized by all members of the group, and all members of the group are required to sign this declaration.

I/We declare that: (i) the assignment here submitted is original except for source material explicitly acknowledged/all members of the group have read and checked that all parts of the piece of work, irrespective of whether they are contributed by individual members or all members as a group, here submitted are original except for source material explicitly acknowledged; (ii) the piece of work, or a part of the piece of work has not been submitted for more than one purpose (e.g. to satisfy the requirements in two different courses) without declaration; and (iii) the submitted soft copy with details listed in the <Submission Details> is identical to the hard copy(ies), if any, which has(have) been / is(are) going to be submitted. I/We also acknowledge that I am/we are aware of the University's policy and regulations on honesty in academic work, and of the disciplinary guidelines and procedures applicable to breaches of such policy and regulations, as contained in the University website http://www.cuhk.edu.hk/policy/academichonesty/.

In the case of a group project, we are aware that all members of the group should be held responsible and liable to disciplinary actions, irrespective of whether he/she has signed the declaration and whether he/she has contributed, directly or indirectly, to the problematic content.

I/we declare that I/we have not distributed/ shared/ copied any teaching materials without the consent of the course teacher(s) to gain unfair academic advantage in the assignment/ course.

I/we declare that I/we have read and understood the University's policy on the use of AI for academic work. I/we confirm that I/we have complied with the instructions given by my/our course teacher(s) regarding the use of AI tools for this assignment and consent to the use of AI content detection software to review my/our submission.

I/We also understand that assignments without a properly signed declaration by the student concerned and in the case of a group project, by all members of the group concerned, will not be graded by the teacher(s).

Shena
Signature(s)

1/2/2025
Date

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Student ID(s)

CSCI 3320
Course code

Fundamental of Machine Learning
Course title

Use of AI tools.

I acknowledge the use of Grok (2026, Feb 1) for content guidance.